

1,2,4-Triazolo-quinazolinones as Effective Antifoulants: Molecular Design, Synthesis, and Biological Evaluation

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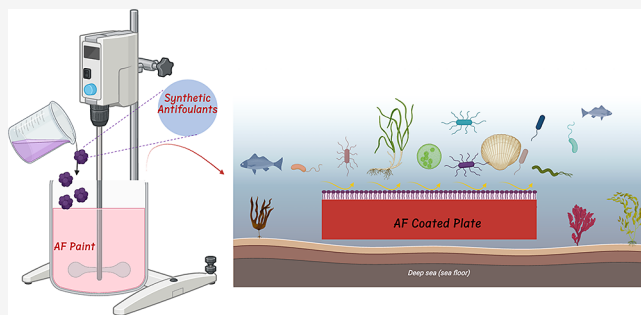


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ABSTRACT: A series of 1,2,4-triazolo-quinazolinones and 1,2-benzisothiazolone derivatives (S1–S12) were successfully synthesized as environmentally friendly alternatives to copper-based antifouling paints using *N*-alkylation, cyclocondensation, and one-pot three-component and amide coupling reactions. The monoclinic structure of single-crystal 1,2,4-triazolo-quinazolin-acetic acid (S8) was confirmed by single-crystal X-ray diffraction analysis. All the synthesized molecules were studied for their *in silico* molecular docking interactions with three target proteins, namely, RbmA, ToxR, and Bap. Following that, the antialgal activity was assessed against two types of marine algae: *Chlorella* sp. and *Chaetoceros curvisetus*. The minimal inhibitory concentration and zone of inhibition have been used to evaluate the antibacterial activities of S1–S12 against both marine Gram-positive (*Staphylococcus aureus*) and Gram-negative (*Vibrio parahaemolyticus* and *Vibrio vulnificus*) bacteria. Additionally, antifouling studies have been done on all the compounds, and among them, 1,2,4-triazolo-quinazolinyl-acetate (S7), 1,2,4-triazolo-quinazolinyl-acetic acid (S8), 1,2,4-triazolo-quinazolinyl-oxobutanoate (S9), benzo[*d*]isothiazolyl butanoate (S10), benzo[*d*]isothiazolyl-acetic acid (S11), and 1,2,4-triazolo-quinazolinyl-acetyl-benzo[*d*]isothiazolone (S12) exhibited good antialgal, antibacterial, and antifouling activities.



INTRODUCTION

Marine biofouling is a complex and undesirable process in which marine micro- and macroorganisms, like bacteria, microalgae, mollusks, seaweeds, barnacles, mussels, and shells, adhere reversibly or irreversibly on hard artificial substrates submerged in the sea.¹ A significant amount of current marine pollution is caused by adhesion, corrosion, and damage from aquatic organisms to submerged infrastructure, such as ships, sonar equipment, and docks, which in turn generates large economic losses.^{2,3} Bacterial communities, which are primary colonizers, initiate biofilm formation by secreting extracellular polymeric substances that enable these bacteria to attach to natural or artificial substrates. Various mechanisms including electrostatic interaction, hydrophobic interaction, hydrogen bonding, Lewis acid–base interaction, and van der Waals interaction are responsible for the adhesion of bacteria.⁴ These microorganisms interact with algal spores, invertebrate larvae, and new microbial colonizers within biofilms and cause biofouling.⁵ In order to prevent this, many modification methods that introduce “nonadhesive” or “antibacterial” properties have been reported in the literature.⁶ Currently, the most popular strategy to prevent biofouling is the application of paints or coatings based on biocides. TBT was the most efficient biocide used in AF paints, and it has been

banned by the International Maritime Organization as it causes significant damage to the marine environment.^{7,8} Nonetheless, cuprous oxide combined with booster biocides continues to rule the antifouling paint/coating industry.⁹ Recently, copper-based paints were also thoroughly reviewed for their pollution of marine ecosystems.¹⁰ This situation shows the critical need for the development of green and nontoxic antifouling agents that will significantly improve marine environments.

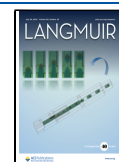
The development and application of natural extracts as green inhibitors have been receiving considerable interest in recent times.^{11,12} However, synthetic methods involve low cost and high yield for the preparation of small molecular chemical inhibitors compared to natural extracts. The majority of 1,2,4-triazoles and their condensed products exhibit strong antiviral, antifungal, and antibacterial properties.^{13,14} Among them, 1,2,4-triazolo[1,5-*a*]pyrimidines and 1,2,4-triazolo[1,5-*a*]pyrimidinones play a significant role in the medicinal

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chemistry. Bülow and Haas initially described the 1,2,4-triazolo[1,5-*a*]pyrimidine (TP) heterocycle in 1909,¹⁵ and one of the TP derivatives was found in the broth of a marine *Streptomyces* sp.¹⁶ Fascinatingly, various TP derivatives such as flumetsulam and similar substances had been developed and marketed as herbicides.¹⁷ The first 1,2,4-triazolo[1,5-*a*]pyrimidine derivative used as a vasodilator and antiplatelet drug was called Trapidil.^{18,19} Triazolo[1,5-*a*]pyrimidines and 1,2,4-triazolo[1,5-*a*]pyrimidinones are two examples of significant drug-like moieties having broad biological applications. These classes of substances have received great attention in agriculture and medicinal chemistry due to their remarkable biological activities in a variety of domains, such as herbicidal,^{20–22} antibacterial,²³ antifungal,²⁴ anticancer,²⁵ antiviral,²⁶ antihypertensive,²⁷ CB2 agonist receptors,²⁸ benzodiazepine receptors,²⁹ and antitubercular agents.³⁰ Quinazolinone compounds are significant categories of heterocyclic compounds with medicinal applications including antibacterial and antifungal activities.³¹ Among the quinazolinone-based compounds, triazolo-quinazolinones show antifungal,³² antimicrobial,³³ antiviral,³⁴ anticancer,³⁵ anticonvulsant,³⁶ anti-inflammatory,³⁷ antioxidant,³⁸ and antihypertensive³⁹ activities. Some biologically active triazolo-quinazolinone/quinazolinone derivatives are listed^{40–43} in Figure 1. 1,2-Benzisothiazol-3(2*H*)-one (BIT) and its analogues also show broad-spectrum antibacterial activity, high stability, and low toxicity and are widely used in the marine industry as antifoulants.^{44,45}

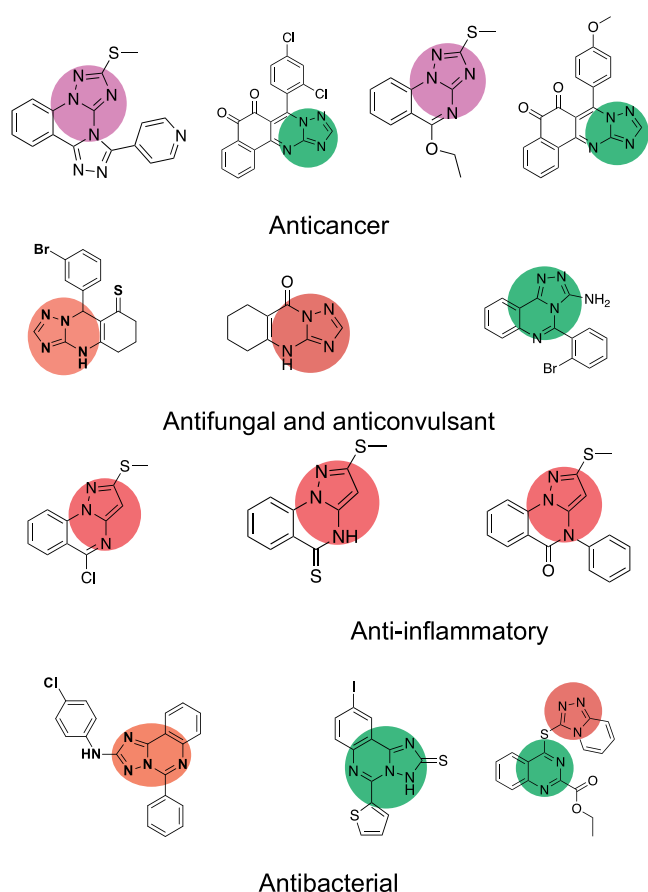


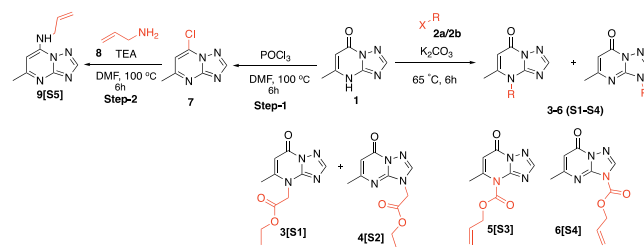
Figure 1. Some biologically active triazolo-quinazolinone/quinazolinone derivatives.

On the other hand, long alkyl chain amines are well soluble in lipids,⁴⁶ altering the permeability of cell membranes and combining with antifouling groups to improve antifouling effectively. In biobased furfurylamines, the chain structure was optimized to provide good toughness and substrate adherence.⁴⁷ Li et al. also synthesized butenolide derivatives with different amine side chains and found that some amine analogues' antilarval settling effect was greatly enhanced by introducing more carbons to their side chains.⁴⁸

Hence, a series of novel 1,2,4-triazolo-pyrimidine/quinazolinone and BIT derivatives with side chains (S1–S12) have been synthesized by using different synthetic methods, and their biological properties have been evaluated. In continuation of our previous work on the preparation of antifouling coatings,⁴⁹ we postulated that simple molecules like triazolopyrimidinones, triazolo-quinazolinones, and 1,2-benzisothiazol-3(2*H*)-ones can be easily synthesized and might exhibit promising antifouling properties.

Herein, we report the synthesis of 1,2,4-triazolo acetates (3,4)/carboxylates (5,6) by *N*-alkylation reaction of 5-methyl 1,2,4-triazolo-pyrimidinone (1) with ethyl bromoacetate (2a) and allyl chloroformate (2b) in the presence of K₂CO₃ (Scheme 1).

Scheme 1. Synthesis of 1,2,4-Triazolo-pyrimidine Derivatives (S1–S5)

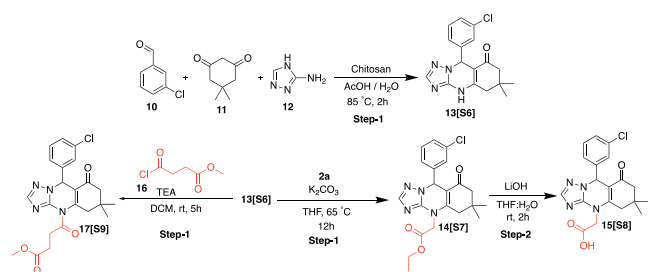


In continuation of our investigation on 1,2,4-triazoles, 7-chloro-5-methyl-[1,2,4]triazolo[1,5-*a*]pyrimidine (7) was prepared by the chlorination of 1 with POCl₃ and further reacted with allylamine (8) to afford *N*-allyl-5-methyl-[1,2,4]triazolo[1,5-*a*]pyrimidin-7-amine (9).

Also, a one-pot, three-component reaction of 3-chlorobenzaldehyde (10), dimedone (11), and 4*H*-1,2,4-triazol-3-amine (12) in the presence of chitosan and aqueous solution of acetic acid afforded 9-(3-chlorophenyl)-6,6-dimethyl-5,6,7,9-tetrahydro-[1,2,4]triazolo[5,1-*b*]quinazolin-8(4*H*)-one (13). This further on *N*-alkylation reaction yielded ethyl 2-(9-(3-chlorophenyl)-6,6-dimethyl-8-oxo-5,7,8,9-tetrahydro-[1,2,4]triazolo[5,1-*b*]quinazolin-4(6*H*)-yl)acetate (14) and followed by ester hydrolysis produced 2-(9-(3-chlorophenyl)-6,6-dimethyl-8-oxo-5,7,8,9-tetrahydro-[1,2,4]triazolo[5,1-*b*]quinazolin-4(6*H*)-yl)acetic acid (15) with good yields. Using 13 as a starting material, methyl 4-(9-(3-chlorophenyl)-6,6-dimethyl-8-oxo-5,7,8,9-tetrahydro-[1,2,4]triazolo[5,1-*b*]quinazolin-4(6*H*)-yl)-4-oxobutanoate (17) was also prepared via *N*-alkylation reaction with methyl succinyl chloride (16), which is prepared from monomethyl succinate (Scheme 2).⁵⁰

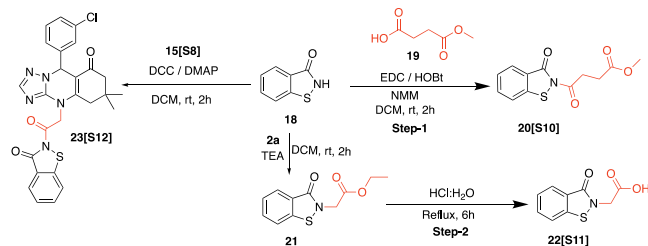
On the other hand, methyl 4-oxo-4-(3-oxobenzod[*d*]-isothiazol-2(3*H*)-yl)butanoate (20) was synthesized by the amidation reaction between benzo[*d*]isothiazol-3(2*H*)-one (BIT) (18) and monomethyl succinate (19). 2-(3-Oxobenzod[*d*]isothiazol-2(3*H*)-yl)acetic acid (22) was also prepared by *N*-alkylation of BIT and followed by the ester hydrolysis of

Scheme 2. Synthesis of 1,2,4-Triazolo-quinazolines (S6–S9)



ethyl 2-(3-oxobenzodithiazol-2(3H)-yl)acetate (21). In the same way, since we are dealing with 1,2,4-triazoles and BIT, we intended to synthesize a hybrid molecule of BIT and 1,2,4-triazoles and assess its activity. The amidation reaction between 15 and 18 gave 2-(2-(9-(3-chlorophenyl)-6,6-dimethyl-8-oxo-5,7,8,9-tetrahydro-[1,2,4]triazolo[5,1-*b*]-quinazolin-4(6H)-yl)acetyl)benzo[*d*]isothiazol-3(2H)-one (23) with good yields (Scheme 3).

Scheme 3. Synthesis of BIT Derivatives (S10–S12)



MATERIALS AND METHODS

General Remarks. All the reactive-grade chemicals, reagents, and solvents were purchased from a commercial source and used without further purification. Ethyl acetoacetate, >99%; 4H-1,2,4-triazol-3-amine, >98%; allylamine, >98%; benzo[*d*]isothiazol-3(2H)-one, >98%; 4-methoxy-4-oxobutanoic acid, >99%; ethyl bromoacetate, >99%; allyl chloroformate, >98%; 3-chlorobenzaldehyde, >98%; 1,3-dimedone, >99%; 1-ethyl-3-(3-dimethylaminopropyl)carbodiimide hydrochloride, >99%; 1-hydroxybenzotriazole, >97%; *N,N'*-dicyclohexylcarbodiimide, >99%; 4-dimethylaminopyridine, >99%; phosphorus(V) oxychloride, >99%; triethylamine, >99.5%; potassium carbonate, >99%; cesium carbonate, >99%; chitosan, >75%; acetic acid, >99%; *N*-methyl morpholine, >99.5%; lithium hydroxide, >98%; hydrochloric acid, 37%; tetrahydrofuran, >99%; *N,N*-dimethylformamide, >99.8%; dichloromethane, >99.8%; ethanol, >95%; and toluene, >99% were purchased from Shanghai Aladdin Bio-Chem Technology Co., Ltd. Distilled water was used. Reactions were carried out in closed glass round-bottom flasks in an oil bath, under magnetic stirring conditions. TLC analysis was conducted using aluminum silica gel plates GF254 (Shanghai Shengya Chemical Co., Ltd.) with an ethyl acetate and petroleum ether mixture as well as a dichloromethane and methanol mixture as the developing solvent. All the final compounds were purified using column chromatography on 200–400 mesh, QingDao HaiYang Chemical Co. Ltd. All the compounds were determined by ^1H and ^{13}C NMR and NOESY. ^1H NMR spectra were recorded at 400 MHz and ^{13}C NMR spectra at 100 MHz in a Bruker Avance 400 spectrometer in $\text{DMSO-}d_6$, CDCl_3 with TMS as the reference.

Preparation of 5-Methyl-[1,2,4]triazolo[1,5-*a*]pyrimidin-7(4H)-one (1). Ethyl acetoacetate (1, 50 g, 384 mmol) and 4H-1,2,4-triazol-3-amine (2, 42 g, 500 mmol) were taken in a 1000 mL round-bottom flask with acetic acid (250 mL), and the mixture was stirred at

130 °C for 12 h. After the completion of the reaction, it was allowed to cool at room temperature and then concentrated under reduced pressure. The resulting solid was filtered, washed with ethyl acetate, and used for the next reactions.

White solid, yield 89% (51 g); ^1H NMR (400 MHz, $\text{DMSO-}d_6$, TMS, δ in ppm): 13.22 (s, 1H, NH), 8.22 (s, 1H, $\text{N}=\text{CH}$), 5.85 (s, 1H, $\text{C}=\text{CH}$), 2.35 (s, 3H, CH_3). ^{13}C NMR (100 MHz, $\text{DMSO-}d_6$, TMS, δ in ppm): 156.27, 152.31, 152.06, 151.02, 98.25, 19.06.

Preparation of 1,2,4-Triazolo-pyrimidinyl-acetate/carboxylates (S1–S4). A dried 250 mL round-bottom flask was equipped with a magnetic stir bar, charged with 5-methyl-[1,2,4]triazolo[1,5-*a*]pyrimidin-7(4H)-one (20 g, 133 mmol) and potassium carbonate (36.8 g, 266 mmol) in THF (100 mL), and stirred at 45 °C for 30 min. Then, ethyl bromoacetate (24.49 g, 146.6 mmol) or allyl chloroformate (17.6 g, 146.6 mmol) was added dropwise with an addition funnel, and the reaction was stirred at 65 °C for 6 h. After the completion of the reaction, the reaction was quenched with sodium bicarbonate solution and the mixture was extracted with ethyl acetate (3 $\times$ 30 mL). The combined organic layers were washed with brine and dried over MgSO_4 . After the organic phase was filtered and concentrated, the residue was purified by column chromatography using petroleum ether and ethyl acetate to produce pure products.

Ethyl 2-(5-Methyl-7-oxo-[1,2,4]triazolo[1,5-*a*]pyrimidin-4(7H)-yl)acetate (S1). White solid, yield 52% (16.50 g); ^1H NMR (400 MHz, $\text{DMSO-}d_6$, TMS, δ in ppm): 8.25 (s, 1H, $\text{N}=\text{CH}$), 6.09 (s, 1H, $\text{C}=\text{CH}$), 5.14 (s, 2H, $\text{CH}_2-\text{C}=\text{O}$), 4.23–4.17 (q, 2H, $\text{CH}_2-\text{C}-\text{O}$), 2.41 (s, 3H, CH_3), 1.22 (t, 3H, $\text{O}-\text{C}-\text{CH}_3$). ^{13}C NMR (100 MHz, $\text{DMSO-}d_6$, TMS, δ in ppm): 167.82, 155.20, 153.27, 152.31, 152.24, 100.69, 62.32, 48.58, 18.66, 14.39.

Ethyl 2-(5-Methyl-7-oxo-[1,2,4]triazolo[1,5-*a*]pyrimidin-3(7H)-yl)acetate (S2). White solid, yield 41% (13 g); ^1H NMR (400 MHz, $\text{DMSO-}d_6$, TMS, δ in ppm): 8.94 (s, 1H, $\text{N}=\text{CH}$), 6.10 (s, 1H, $\text{C}=\text{CH}$), 5.10 (s, 2H, $\text{CH}_2-\text{C}=\text{O}$), 4.32–4.26 (q, 2H, $\text{CH}_2-\text{C}-\text{O}$), 2.36 (s, 3H, $-\text{CH}_3$), 1.31 (t, 3H, $\text{O}-\text{C}-\text{CH}_3$). ^{13}C NMR (100 MHz, $\text{DMSO-}d_6$, TMS, δ in ppm): 167.31, 164.52, 155.86, 148.66, 143.59, 102.34, 62.31, 44.79, 24.22, 14.42.

Allyl 5-Methyl-7-oxo-[1,2,4]triazolo[1,5-*a*]pyrimidine-4(7H)-carboxylate (S3). Pale yellow solid, yield 60% (19 g); ^1H NMR (400 MHz, $\text{DMSO-}d_6$, TMS, δ in ppm): 8.23 (s, 1H, $\text{N}=\text{CH}$), 6.03 (s, 1H, $\text{C}=\text{CH}$), 5.97–6.02 (m, 1H, $\text{CH}=\text{CH}$), 5.22 (d, $J = 8 \text{ Hz}$, 1H, $\text{CHH}=\text{CH}$), 5.08 (d, $J = 16 \text{ Hz}$, 1H, $\text{CHH}=\text{CH}$), 4.89 (d, $J = 4 \text{ Hz}$, 2H, CH_2), 2.43 (s, 3H, $-\text{CH}_3$). ^{13}C NMR (100 MHz, $\text{DMSO-}d_6$, TMS, δ in ppm): 155.42, 153.11, 152.30, 152.26, 152.21, 132.14, 117.66, 100.61, 49.70, 18.46.

Allyl 5-Methyl-7-oxo-[1,2,4]triazolo[1,5-*a*]pyrimidine-3(7H)-carboxylate (S4). Pale yellow solid, yield 30% (9.50 g); ^1H NMR (400 MHz, $\text{DMSO-}d_6$, TMS, δ in ppm): 8.87 (s, 1H, $\text{N}=\text{CH}$), 6.00–6.08 (m, 1H, $\text{CH}=\text{CH}$), 5.98 (s, 1H, $\text{C}=\text{CH}$), 5.28 (d, $J = 12 \text{ Hz}$, 1H, $\text{CHH}=\text{CH}$), 5.22 (d, $J = 16 \text{ Hz}$, 1H, $\text{CHH}=\text{CH}$), 4.69 (d, $J = 4 \text{ Hz}$, 2H, CH_2), 2.28 (s, 3H, $-\text{CH}_3$). ^{13}C NMR (100 MHz, $\text{DMSO-}d_6$, TMS, δ in ppm): 164.40, 156.06, 148.65, 143.10, 131.95, 119.26, 101.88, 45.92, 24.30.

Preparation of *N*-Allyl-5-methyl-[1,2,4]triazolo[1,5-*a*]pyrimidin-7-amine (S5). 5-Methyl-[1,2,4]triazolo[1,5-*a*]pyrimidin-7(4H)-one (15 g, 100 mmol) and DMF (75 mL) were taken in a 250 mL round-bottom flask, and then POCl_3 (14 mL, 150 mmol) was added followed by stirring at 100 °C for 6 h. The resultant oil was poured into ice-cold water and extracted with ethyl acetate (3 $\times$ 30 mL). The combined organic layers were washed with saturated NaHCO_3 solution, dried over MgSO_4 , and concentrated. The resulting crude compound (13.0 g, 77.3 mmol), TEA (21.7 mL, 154.7 mmol), and DMF (55 mL) were taken in a 250 mL round-bottom flask and stirred at room temperature for 30 min. After that, allylamine (4.85 g, 85.11 mmol) was added dropwise followed by stirring at 100 °C, for 6 h. After completion of the reaction, the reaction was quenched with ice-cold water and the aqueous layer was extracted with ethyl acetate (3 $\times$ 30 mL). The combined organic layers were dried over MgSO_4 and concentrated. The resulting crude product was purified by column chromatography using petroleum ether and ethyl acetate to produce a pure product.

Yellow solid, yield 61% (9.0 g); ^1H NMR (400 MHz, $\text{DMSO}-d_6$, TMS, δ in ppm): 8.42 (s, 1H, NH), 8.39 (s, 1H, $\text{N}=\text{CH}$), 6.25 (s, 1H, $\text{C}=\text{CH}$), 5.95–5.86 (m, 1H, $\text{CH}=\text{N}$), 5.25 (d, $J = 16$ Hz, 1H, $\text{CHH}=\text{N}$), 5.16 (d, $J = 12$ Hz, 1H, $\text{CHH}=\text{N}$), 4.03 (t, 1H, CH_2), 2.43 (s, 3H, $-\text{CH}_3$). ^{13}C NMR (100 MHz, $\text{DMSO}-d_6$, TMS, δ in ppm): 164.15, 155.75, 154.64, 147.66, 147.66, 133.96, 116.78, 88.47, 44.07, 25.24.

Preparation of 9-(3-Chlorophenyl)-6,6-dimethyl-5,6,7,9-tetrahydro-[1,2,4]triazolo[5,1-b]quinazolin-8(4H)-one (S6). A dried 500 mL round-bottom flask was equipped with a magnetic stir bar and a water condenser charged with 200 mL of an aqueous solution of 2% acetic acid and chitosan (1.09 g, 0.714 mmol). Then, 3-chlorobenzaldehyde (10.0 g, 71.4 mmol), 4H-1,2,4-triazol-3-amine (10.8 g, 128.5 mmol), and dimedone (11 g, 78.5 mmol) were added and stirred at 85 °C for 2 h. After completion of the reaction, the obtained precipitate was filtered and dried.

Yellow solid, yield 90% (28 g); ^1H NMR (400 MHz, $\text{DMSO}-d_6$, TMS, δ in ppm): 11.23 (s, 1H, $-\text{NH}$), 7.73 (s, 1H, $\text{CH}=\text{N}$), 7.35–7.28 (m, 4H, $\text{Ar}-\text{H}$), 6.25 (s, 1H, CH), 2.57 (s, 2H, CH_2), 2.23 (d, $J = 16$ Hz, CH_2), 2.12 (d, $J = 16$ Hz, CH_2), 1.06 (s, 3H, CH_3), 0.98 (s, 3H, CH_3). ^{13}C NMR (100 MHz, $\text{DMSO}-d_6$, TMS, δ in ppm): 193.54, 151.27, 150.76, 147.24, 144.31, 133.31, 130.82, 128.24, 127.56, 126.11, 105.42, 57.97, 50.22, 32.74, 28.81, 27.40.

Preparation of Ethyl 2-(9-(3-Chlorophenyl)-6,6-dimethyl-8-oxo-5,7,8,9-tetrahydro-[1,2,4]triazolo[5,1-b]quinazolin-4(6H)-yl)acetate (S7). In a dried 250 mL round-bottom flask, 9-(3-chlorophenyl)-6,6-dimethyl-5,6,7,9-tetrahydro-[1,2,4]triazolo[5,1-b]quinazolin-8(4H)-one (12.0 g, 36.5 mmol) and potassium carbonate (10 g, 73 mmol) were added in THF (50 mL). Furthermore, ethyl bromoacetate (4.42 mL, 40.2 mmol) was diluted in THF and added dropwise to the reaction mixture, and the reaction was continued at 65 °C for 12 h. Then, the reaction was quenched with saturated NaHCO_3 solution, and the mixture was extracted in ethyl acetate (4 $\times$ 10 mL). The combined organic layers were dried over MgSO_4 and concentrated. The resulting crude product was purified by column chromatography using CH_2Cl_2 and MeOH.

Pale yellow solid, yield 85% (13 g); ^1H NMR (400 MHz, $\text{DMSO}-d_6$, TMS, δ in ppm): 7.76 (s, 1H, $\text{CH}=\text{N}$), 7.36–7.24 (m, 4H, $\text{Ar}-\text{H}$), 6.31 (s, 1H, CH), 4.99–4.88 (m, 2H, $\text{N}-\text{CH}_2$), 4.25–4.20 (m, 2H, $\text{O}-\text{CH}_2$), 2.71–2.56 (m, 2H, CH_2), 2.26 (d, $J = 16$ Hz, CH_2), 2.13 (d, $J = 16$ Hz, CH_2), 1.24 (t, 3H, CH_2-CH_3), 1.06 (s, 3H, $-\text{CH}_3$), 0.95 (s, 3H, $-\text{CH}_3$).

Preparation of 2-(9-(3-Chlorophenyl)-6,6-dimethyl-8-oxo-5,7,8,9-tetrahydro-[1,2,4]triazolo[5,1-b]quinazolin-4(6H)-yl)acetic Acid (S8). The dried 100 mL round-bottom flask was charged with a THF/ H_2O (25 mL:25 mL) solvent mixture followed by the addition of ethyl 2-(9-(3-chlorophenyl)-6,6-dimethyl-8-oxo-5,7,8,9-tetrahydro-[1,2,4]triazolo[5,1-b]quinazolin-4(6H)-yl)acetate (10 g, 24.15 mmol) and LiOH (1.15 g, 48.3 mmol) to the reaction mixture at 0 °C and stirring at room temperature for 2 h. Eventually, the reaction mixture was acidified with diluted HCl up to the pH value 3–4 and extracted in CH_2Cl_2 (3 $\times$ 10 mL). The combined organic layers were dried over MgSO_4 and concentrated to get the final product.

Pale yellow solid, yield 96% (9.0 g); ^1H NMR (400 MHz, $\text{DMSO}-d_6$, TMS, δ in ppm): 13.49 (s, 1H, $-\text{OH}$), 7.76 (s, 1H, $\text{CH}=\text{N}$), 7.4–7.27 (m, 4H, $\text{Ar}-\text{H}$), 6.29 (s, 1H, CH), 4.91–4.78 (m, 2H, CH_2), 2.70–2.55 (m, 2H, CH_2), 2.26 (d, $J = 16$ Hz, CH_2), 2.13 (d, $J = 16$ Hz, CH_2), 1.06 (s, 3H, $-\text{CH}_3$), 0.95 (s, 3H, $-\text{CH}_3$). ^{13}C NMR (100 MHz, $\text{DMSO}-d_6$, TMS, δ in ppm): 193.96, 170.48, 151.61, 150.53, 148.86, 144.07, 133.47, 130.62, 128.38, 127.64, 126.40, 107.73, 57.51, 49.50, 46.98, 38.09, 32.47, 29.03, 27.26.

Preparation of Methyl 4-(9-(3-Chlorophenyl)-6,6-dimethyl-8-oxo-5,7,8,9-tetrahydro-[1,2,4]triazolo[5,1-b]quinazolin-4(6H)-yl)-4-oxobutanoate (S9). Methyl succinyl chloride was prepared by the chlorination of monomethyl succinate as mentioned in the previous report. A 250 mL dried round-bottom flask tightly fitted with a nitrogen gas and water condenser was taken followed by the addition of monomethyl succinate (19.75 g, 150 mmol) and thionyl chloride (54 mL). The reaction was stirred at 80 °C for 2 h, and the TLC was checked to monitor the reaction. Finally, the reaction mixture was

concentrated in vacuo to remove the excess thionyl chloride and the resulting oil was used for further reaction. In the next step, 9-(3-chlorophenyl)-6,6-dimethyl-5,6,7,9-tetrahydro-[1,2,4]triazolo[5,1-b]quinazolin-8(4H)-one (10 g, 30.48 mmol) in DCM (50 mL) and TEA (8.5 mL, 61 mmol) were taken in a 250 mL dried round-bottom flask and stirred at room temperature for 5 h. After that, methyl succinyl chloride (5.48 g, 36.5 mmol) was added dropwise and the reaction was continued for 5 h. The reaction was monitored by examining the TLC, the reaction was quenched with saturated NaHCO_3 solution, and the mixture was extracted in CH_2Cl_2 (3 $\times$ 20 mL). The combined organic layers were dried over MgSO_4 and concentrated. The resulting crude product was purified by column chromatography using CH_2Cl_2 and MeOH.

White solid, yield 74% (10 g); ^1H NMR (400 MHz, $\text{DMSO}-d_6$, TMS, δ in ppm): 7.76 (s, 1H, $\text{CH}=\text{N}$), 7.36–7.25 (m, 4H, $\text{Ar}-\text{H}$), 6.31 (s, 1H, CH), 4.94 (t, 2H, CH_2), 4.23 (t, 2H, CH_2), 2.71–2.56 (m, 2H, CH_2), 2.26 (d, $J = 16$ Hz, CH_2), 2.13 (d, $J = 16$ Hz, CH_2), 1.24 (t, 3H, $-\text{CH}_3$), 1.06 (s, 3H, $-\text{CH}_3$), 0.95 (s, 3H, $-\text{CH}_3$).

Preparation of Methyl 4-Oxo-4-(3-oxobenzo[d]isothiazol-2(3H)-yl)butanoate (S10). In a dried 250 mL round-bottom flask, BIT (10 g, 66.22 mmol), 4-methoxy-4-oxobutanoic acid (9.61 g, 72.84 mmol), EDC (13.98 g, 72.84 mmol), and HOBt (9.83 g, 72.84 mmol) were taken in DCM (50 mL) and stirred at 0 °C for 15 min. After that, NMM (14.5 mL, 132.0 mmol) was added to the reaction mixture and the reaction was carried out at room temperature for 2 h. The reaction was monitored by checking the TLC. After the completion of the reaction, the reaction was quenched with ice-cold water and the aqueous layer was extracted with CH_2Cl_2 (3 $\times$ 30 mL). The combined organic layers were dried over MgSO_4 and concentrated. The resulting crude product was purified by column chromatography using petroleum ether and ethyl acetate to produce a pure product.

Yellow solid, yield 88% (15.50 g); ^1H NMR (400 MHz, CDCl_3 , TMS, δ in ppm): 8.03 (d, $J = 8$ Hz, 1H, $\text{Ar}-\text{H}$), 7.70 (t, 1H, $\text{Ar}-\text{H}$), 7.52 (d, $J = 8$ Hz, 1H, $\text{Ar}-\text{H}$), 7.41 (t, 1H, $\text{Ar}-\text{H}$), 3.72 (s, 3H, $\text{O}-\text{CH}_3$), 3.50 (t, 2H, $-\text{CH}_2-$), 2.79 (t, 2H, $-\text{CH}_2-$). ^{13}C NMR (100 MHz, $\text{DMSO}-d_6$, TMS, δ in ppm): 172.95, 172.47, 163.73, 141.40, 135.13, 127.51, 126.68, 125.52, 122.63, 52.01, 32.82, 28.24.

Preparation of 2-(3-Oxobenzo[d]isothiazol-2(3H)-yl)acetic Acid (S11). In a dried 250 mL round-bottom flask, BIT (15 g, 99.33 mmol) and ethyl bromoacetate (13 mL, 119.2 mmol) were taken in DCM (75 mL) and stirred at room temperature for 2 h. After the completion of the reaction, the resulting solid was filtered and washed with petroleum ether. The resulting solid intermediate compound (14.0 g, 59.0 mmol) was taken in a 100 mL round-bottom flask, and HCl (37%) and Milli-Q H_2O (3.5 and 3.5 mL) were added and refluxed for 6 h. Finally, the resulting white solid was filtered and dried.

White solid, yield 85% (11.50 g); ^1H NMR (400 MHz, $\text{DMSO}-d_6$, TMS, δ in ppm): 13.14 (s, 1H, $-\text{OH}$), 8.0 (d, $J = 8$ Hz, 1H, $\text{Ar}-\text{H}$), 7.91 (d, $J = 8$ Hz, 1H, $\text{Ar}-\text{H}$), 7.71 (t, 1H, $\text{Ar}-\text{H}$), 7.46 (t, 1H, $\text{Ar}-\text{H}$), 4.61 (s, 2H, CH_2). ^{13}C NMR (100 MHz, $\text{DMSO}-d_6$, TMS, δ in ppm): 169.86, 165.36, 141.78, 132.55, 126.13, 125.93, 123.75, 122.34, 44.75.

Preparation of 2-(2-(9-(3-Chlorophenyl)-6,6-dimethyl-8-oxo-5,7,8,9-tetrahydro-[1,2,4]triazolo[5,1-b]quinazolin-4(6H)-yl)acetyl)-benzo[d]isothiazol-3(2H)-one (S12). A 250 mL dried round-bottom flask was charged with DCM (50 mL), 2-(9-(3-chlorophenyl)-6,6-dimethyl-8-oxo-5,7,8,9-tetrahydro-[1,2,4]triazolo[5,1-b]quinazolin-4(6H)-yl)acetic acid (10 g, 25.90 mmol), BIT (3.91 g, 25.90 mmol), and DCC (6.4 g, 31 mmol) at 0 °C and stirred for 15 min. Subsequently, 4-DMAP (632 mg, 5.1 mmol) was added to the reaction mixture and the reaction was continued for 2 h. At the end, the reaction was quenched with brine and the mixture was extracted in ethyl acetate (3 $\times$ 20 mL). The combined organic layers were dried over MgSO_4 and concentrated. The resulting crude product was purified by column chromatography using CH_2Cl_2 and MeOH.

Yellow solid, yield 67% (9.0 g); ^1H NMR (400 MHz, $\text{DMSO}-d_6$, TMS, δ in ppm): 8.06 (m, 2H, $\text{Ar}-\text{H}$), 7.85 (t, 1H, $\text{Ar}-\text{H}$), 7.74 (s, 1H, $\text{CH}=\text{N}$), 7.56–7.47 (m, 2H, $\text{Ar}-\text{H}$), 7.39–7.32 (m, 3H, $\text{Ar}-\text{H}$), 6.32 (s, 1H, CH), 5.70–5.53 (m, 2H, CH_2), 2.81–2.59 (m, 2H, CH_2), 2.27 (d, $J = 16$ Hz, CH_2), 2.13 (d, $J = 16$ Hz, CH_2), 1.05 (s, 3H,

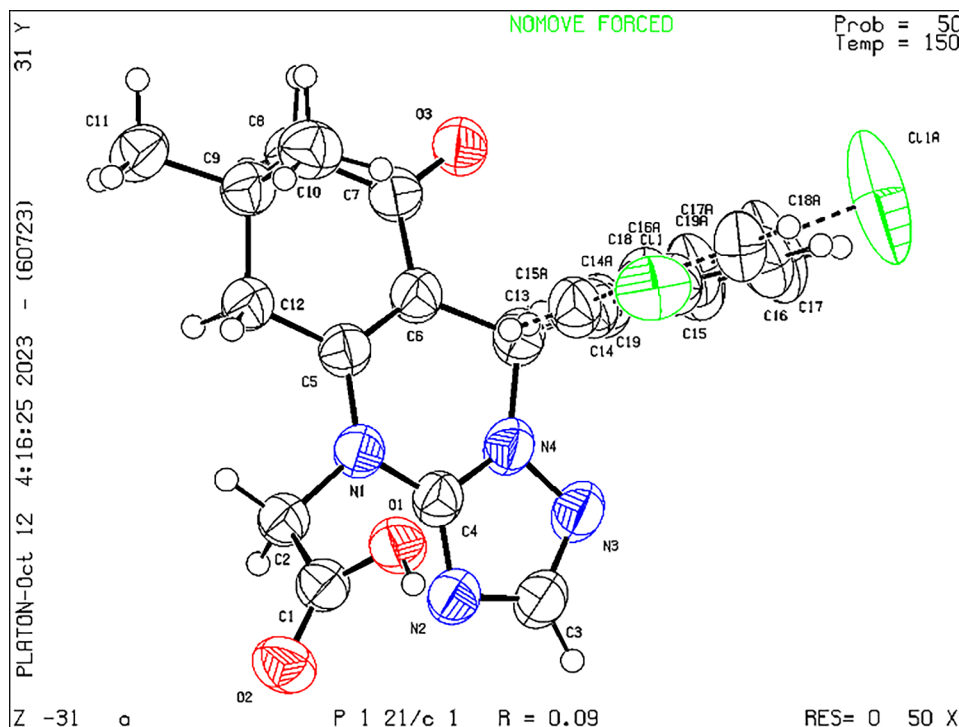


Figure 2. PLATON diagram of 1,2,4-triazolo-quinazolinyl-acetic acid (S8).

$-\text{CH}_3$), 0.94 (s, 3H, $-\text{CH}_3$). ^{13}C NMR (100 MHz, $\text{DMSO}-d_6$, TMS, δ in ppm): 194.02, 169.48, 151.41, 150.57, 148.67, 143.96, 133.44, 130.72, 128.43, 127.58, 126.31, 125.31, 124.57, 121.97, 107.92, 57.46, 53.14, 49.49, 46.81, 37.99, 32.51, 28.95, 27.25.

All the desired products were successfully synthesized on a gram scale and obtained in good yields.

In Silico Studies. Molecular Docking Analysis. The maestro visualization tool (Schrödinger Release: Maestro, version 10.5, Schrödinger, LLC, NY 2016–1) was used to display receptors, ligand structures, receptor–ligand interactions, and their hydrogen bonding network to generate the images. The synthesized compounds were studied for their in silico molecular docking interactions for three target proteins, namely, RbmA, ToxR, and Bap. Target receptors used in this study are those virulent proteins that are significant for bacteria's biofilm formation and survival by attachment which pose as essential targets for antifouling agents. The most commonly associated biofouling *Vibrio* species are *Vibrio cholerae*, *Vibrio parahaemolyticus*, and *Vibrio vulnificus*. The target sequence of the functional analogue was downloaded from the Uniprot database (www.uniprot.org, ID: A0A233HBR5). The structural model was generated using the PHYRE server and validated with the SWISS model. X-ray crystallographic structures were obtained from the protein databank (<https://www.rcsb.org/>) (see Supplementary data). The protein preparation wizard was used to prepare all the proteins used in docking. The preparation included the assignment of hydrogen bonds, bond orders, hydrogen addition, optimization, protein minimization, and deletion of water molecules beyond the het group. Sitemap tool analysis was used to identify highly potential binding sites of ligands on proteins. For the preparation of ligands, the Ligprep application was employed (v3.7, Schrödinger Release 2016–1). A protein receptor grid was generated using the Glide application (assignment of ligand binding sites for docking). Furthermore, all ligands were docked using Glide's ligand docking module. The Glide score (G score) was visualized using extra-precision (XP-visualizer module).

In Vitro Studies. Algal Inhibition Measurements. The two algae, namely, *Chlorella* sp. and *Chaetoceros curvisetus*, were procured from the algal culture group, Ocean College of Hainan University. *Chlorella* sp. and *C. curvisetus* extracts were diluted with seawater to a certain

concentration. The concentration of the algal solution was measured three times to obtain an average concentration for three consecutive days. The difference in absorbance values of the algal solution was measured, and the effects of 12 compounds in the two algal solutions were analyzed using a UV–visible spectrophotometer (TU-1901) at 685 nm. The pure algal solution was taken as blank and the algal solution containing DMSO as the control. The obtained data were statistically analyzed by using Origin 8.6 software (Origin Lab, Northampton, MA, USA).

Antibacterial Studies. Antibacterial activity has been assessed for a range of marine Gram-positive (*Staphylococcus aureus*) and Gram-negative (*Vibrio parahaemolyticus* and *Vibrio vulnificus*) bacteria by calculating the minimum inhibitory concentration (MIC) using 96-well microtiter plates and the broth microdilution method by preparing concentrations of 0, 2, 4, 8, 16, 32, 64, 128, and 256 $\mu\text{g}/\text{mL}$. In addition, the disk diffusion experiment was also performed using Mueller–Hinton agar plates. The pathogenic bacteria were subcultured in Luria–Bertani (LB) broth and allowed to grow under optimal conditions for the whole night at 37 $^\circ\text{C}$. The agar plates were coated with the bacterial cultures set to 0.5 McFarland standards (1×10^8 CFU/mL), and the plates were incubated for 24 h at 37 $^\circ\text{C}$. The experiment was carried out in triplicate, and the zone of inhibition (mm) was calculated. The reference bacterial strains used for both assays are from MTCC = Microbial Type Culture Collection, Hainan University, Haikou, China.

SEM Analysis of Antibacterial Activity. In summary, 50 $\mu\text{g}/\text{mL}$ of synthesized compounds were added to 100 μL of log-phase bacterial cultures ($\sim 10^8$ CFU/mL) and incubated for three hours at 37 $^\circ\text{C}$. It was then centrifuged for 10 min at $3000 \times g$. After harvesting the pellets, 10 mL of $1 \times$ phosphate-buffered saline (PBS) solution (pH 7.4) was used to resuspend them, followed by three subsequent PBS buffer washes to remove the excess media. Afterward, 500 μL of 2.5% glutaraldehyde solution in 0.1 M Tris-buffered saline (TBS) was added to the pellets, and they were kept at 37 $^\circ\text{C}$ for 30 min. Again, the pellets were washed and resuspended in 0.1 M TBS solution. Next, a coverslip coated with poly-L-lysine was coated with 50 μL of the bacterial suspension. To create clear micrographs, the samples were sputter coated with a double-sided metal stub that had a 10 nm gold (Au) thickness after being air-dried using pure ethanol. Lastly,

pictures were taken and the samples were examined with an SEM analyzer.

Antifouling Field Tests. In order to evaluate the potential antifouling properties of synthesized compounds, zinc acrylic-based antifouling paint was formulated as follows: zinc acrylic resin (laboratory homemade), 75 g; zinc oxide (Tianjin Yong Da Chemical Reagent Development Center), 5 g; bentonite (Shanghai Macklin Biochemical Co., Ltd.), 0.2 g; colophony (Shanghai Macklin Biochemical Co., Ltd.), 5 g; ferric sesquioxide (Shanghai Aladdin Biochemical Technology Co., Ltd.), 5 g; anino-2-methyl-1-propanol (Shanghai Macklin Biochemical Co., Ltd.) 0.05 g; antifouling agent, 10 g; xylene (Shanghai Aladdin Biochemical Technology Co., Ltd.); and *n*-butanol (Shanghai Aladdin Biochemical Technology Co., Ltd.) mixed solvent (5 mL). The antifouling agent free paint or control paint (CP) was prepared at a laboratory scale. The preparation method is to put all the reagents except the antifouling agent into the high-speed disperser in proportion and grind them with the ball grinding method for 25 min. After preparation of CPs, the synthesized antifouling agents were also mixed with other reagents in the high-speed disperser for 25 min. Both butenolide paint (BP) and Cu₂O paints were also prepared as reference paints. In all cases, the synthesized antifouling agents were tested at 10% w/w. Antifouling paints were tested against a negative control (CP) at the South China Sea (110.190E, 20.10N) from November to December in China at temperatures ranging from 22 to 28 °C. Epoxy resin plates (200 mm × 300 mm) were used for antifouling coatings after being cleaned with ethanol and deionized water. Each coated plate, including the control plate (CP), was dried at room temperature before being submerged in marine water between one and two meters below the surface of the ocean. After 30 and 60 days, the plates were taken out, sludge was cleaned with seawater, photos were taken, and they were promptly kept for further marine field tests.

RESULTS AND DISCUSSION

The novel chemical structure and interesting biological properties of 1,2,4-triazolo-quinazolinones have attracted a lot of interest. There is hope for the discovery of novel compounds with good biological activities, enhanced efficacy, and environmental sustainability if research continues on their synthesis.

Single-Crystal X-ray Diffraction Study. The single crystals of 1,2,4-triazolo-quinazolinyl-acetic acid (S8) were prepared by dissolving the compound in dichloromethane and ethanol. The structure of the grown single crystal was confirmed by single-crystal X-ray diffraction analysis. The crystal is monoclinic, and the estimated lattice parameters are predicted to be unit cell dimensions $a = 14.8967(5)$ Å, $b = 9.3991(3)$ Å, $c = 15.0054(5)$ Å, volume = 1825.28(11) Å³, $Z = 4$, $T = 150.00$ K, $\mu(\text{CuK}\alpha) = 2.096$ mm⁻¹, and $D_{\text{calc}} = 1.408$ g/cm³. The crystal exhibits a monoclinic structure with a $P2_1/c$ space group, and the molecular structure is shown as a PLATON diagram in Figure 2.

The bond lengths of O–C are O₁–C₁ = 1.322(3) Å, O₂–C₁ = 1.202(3) Å, and O₃–C₇ = 1.237(3) Å, and bond angles are O₁–C₁–C₂ = 114.6(2)°, O₂–C₁–O₁ = 124.8(2)°, O₂–C₁–C₂ = 120.6(2)°, O₃–C₇–C₆ = 120.1(2)°, and O₃–C₇–C₈ = 121.8(2)°. The torsion angles are O₁–C₁–C₂–N₁ = 28.0(3)°, O₂–C₁–C₂–N₁ = –154.6(2)°, and O₃–C₇–C₈–C₉ = –148.6(3)°.

Based on some natural antifoulant structures and previously reported literature studies, we came to know that the alkyl side chain had a notable effect on antilarval settlement activity.^{48,51} Therefore, we wanted to synthesize some 1,2,4-triazolo derivatives and BIT derivatives. In that, the starting material 5-methyl-[1,2,4]triazolo[1,5-*a*]pyrimidin-7(4*H*)-one was synthesized by the cyclocondensation reaction of 4*H*-1,2,4-triazol-

3-amine and ethyl acetoacetate as mentioned in the literature.⁵² There were two alkylated products with yields of 52 and 41% obtained from the *N*-alkylation reaction with ethyl bromoacetate. Conversely, allyl compounds have been obtained with 60 and 30% yields during the reaction of allyl chloroformate. Both of the products were isolated separately using column chromatography and analyzed spectral data. In continuation, chlorination of compound 1 has been done using the reported method.⁵³ We continued the same base K₂CO₃ for the allyamine reaction, but we got fewer yields when compared with TEA. There were many reports on synthesis of 1,2,4-triazolo-quinazolinones using various catalysts.^{54,55} Nevertheless, we have used chitosan as a green catalyst in the presence of an aqueous solution of 2% acetic acid and the yield was about 90%. Therefore, we continued with the larger scale preparation and then used it for the *N*-alkylation followed by the ester hydrolysis reactions. According to the reported literature,⁵⁰ monomethyl succinate was used to prepare the succinyl chloride and used in the *N*-alkylation reaction with BIT, but the yields were lesser than those for the amidation reaction with monomethyl succinate. Therefore, we proceeded with the amidation reaction to get more yields. In keeping with the BIT analogue synthesis, BIT ester was also prepared as mentioned in the literature.⁵⁶ To prepare the BIT acid from ester using an ester hydrolysis reaction, we tried different bases like NaOH, KOH, and LiOH, but none of them gave good yields. Hence, we tried the reaction in an acidic medium (HCl/H₂O) under reflux conditions and received good yields up to 85%.

Finally, we assumed that the quinazolinone ring in S8 and the isothiazolone ring in BIT are responsible for more activities. Hence, we tried to synthesize a hybrid molecule with BIT and 1,2,4-triazolo-quinazolinone acetic acid using amidation reaction using DCC/4-DMAP conditions and we succeeded with 67% yields. We changed the reaction conditions to HATU/DIPEA and EDCI/HOBt to improve the yields. However, the DCC/4-DMAP reaction condition was the best one in terms of yields and reaction time. After synthesizing all the compounds, we confirmed the structures using spectral data and continued the biological analysis.

In Silico Molecular Docking Analysis. In this study, 12 compounds were subjected to molecular docking, among which compounds S7, S8, S9, and S12 showed good results with all three target receptor proteins. The first target protein is ToxR, which regulates virulence factors in *Vibrio parahaemolyticus*, including the production of thermostable direct hemolysin (TDH) and TDH-related hemolysin (TRH), both of which contribute significantly to the bacterium's pathogenicity.⁵⁷ ToxR also regulates other factors such as colonization and stress response genes, allowing *Vibrio parahaemolyticus* to adapt and thrive in various habitats. The second target Bap (Biofilm-associated protein) is a large, multidomain surface protein that is essential for biofilm formation. The *bap* gene encodes the Bap protein, which is frequently seen in strains of *Staphylococcus aureus* and is responsible for increasing bacterial adhesion and accumulation, resulting in the formation of strong biofilms.⁵⁸ The third target is RbmA, a protein controlled by the vibrio polysaccharide system as it plays a crucial role in regulating the production of exopolysaccharide (EPS), a key component of the biofilm matrix.⁵⁹ The result of docking interaction is in the form of binding energies. The lower the binding energy, the stronger the interaction. The

molecular docking interactions for three target proteins are shown in the table (see the Supplementary data).

Compound S7 exhibited good docking scores across all three proteins with -5.414 kcal/mol for RbmA, -3.134 kcal/mol for ToxR, and -4.292 kcal/mol for Bap. S8 also showed promising interactions with scores of -5.941 kcal/mol for RbmA, -4.875 kcal/mol for ToxR, and -5.251 kcal/mol for Bap. S9 demonstrated strong interactions with Bap scoring -5.573 kcal/mol, -6.414 kcal/mol with RbmA, and -4.276 kcal/mol with ToxR. Additionally, S12 exhibits good interactions with RbmA Bap ToxR proteins with docking scores of -5.064 kcal/mol, -5.517 kcal/mol, and -3.733 kcal/mol and forms two hydrogen bonds with Thr267 and Lys 128 respectively. Figure 3 illustrates the docking interactions and hydrogen bonds formed by compounds S9 and S12.

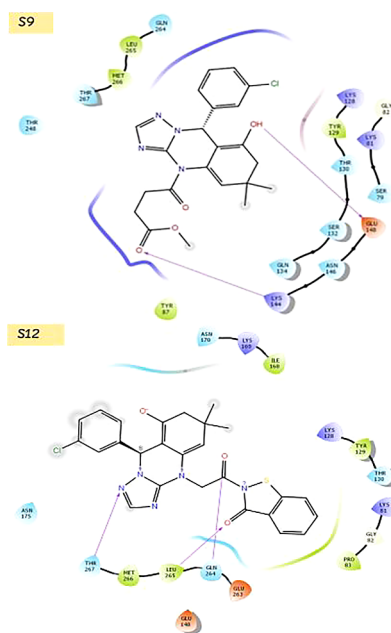


Figure 3. Interactions of Bap with compounds S9 and S12.

The molecular docking results indicate that compounds S7, S8, S11, and S12 effectively interact with the target proteins RbmA, ToxR, and Bap, suggesting their potential as antifouling agents. The formation of hydrogen bonds by S7 with key residues in RbmA supports its role in disrupting biofilm structural integrity. Compounds S9 and S12, with strong interactions with Bap, highlight their capability to hinder bacterial adhesion and aggregation. These findings support the strategic targeting of RbmA, ToxR, and Bap in developing effective antifouling treatments offering a promising approach to mitigating marine biofouling. Further studies are warranted to validate these compounds' efficacy and environmental safety.

Biological Evaluation. Algae and bacteria often form symbiotic relationships and compete for resources within the biofilm layer. This layer provides a substrate for the attachment of larvae and spores of other fouling organisms facilitating their settlement and growth on submerged surfaces like ship hulls, docks, and underwater structures. Hence, we have evaluated all the synthesized compounds for their antialgal, antibacterial, and antifouling activities.

Algal Inhibition Studies. All the synthesized compounds were tested for their algal inhibition activities against two algae,

namely, *Chlorella* and *C. curvisetus*. The highest results in *Chlorella* testing were found with 2 mg/mL of compounds S1, S7, and S9, 6 mg/mL of compound S10, and 10 mg/mL of compounds S8, S11, and S12.

Nevertheless, the most effective doses against *C. curvisetus* were 4 mg/mL for compounds S6, S7, S8, and S9 and 2 mg/mL and 6 mg/mL for compounds S11 and S12, respectively. All the results were analyzed by using Origin 8.6 software and are shown in Figure 4.

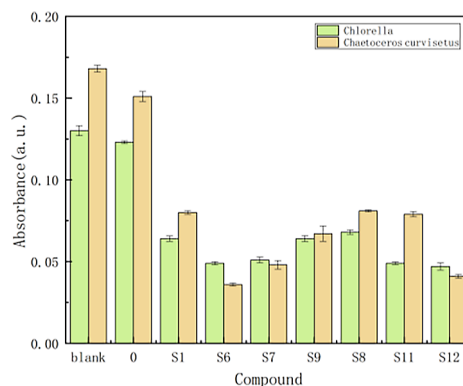


Figure 4. Algal inhibition activity of tested compounds against *Chlorella* and *C. curvisetus*. Values are mean $\pm$ SD, $N = 3$.

Antibacterial Studies. The results of antibacterial activity using both minimum inhibitory concentration (MIC) and zone of inhibition (in mm) methods indicate the efficacy of the synthesized compounds. Each of the compounds was tested separately at a concentration of $50 \mu\text{g}/\text{well}$ against three different bacterial strains: *S. aureus*, *V. parahaemolyticus*, and *V. vulnificus*.

The effectiveness of the drugs to inhibit bacterial growth differed significantly among them. Compounds S3, S6, and S8 showed good antibacterial activity against *V. vulnificus*. At the same time, S3, S4, and S6 showed good activity against *V. parahaemolyticus*. Particularly, 1,2,4-triazolo-oxobutanoate (S9), BIT butanoate (S10), and 1,2,4-triazolo-benzo[d]isothiazolone (S12) exhibited significant antibacterial activity against all strains tested (Figure 5). These results can be corroborated with the results of 1,2,4-triazolo-quinazolinones synthesized by Patel⁶⁰ and quinoline derivatives synthesized by Lavanya.⁶¹ SEM examination revealed that at a maximum concentration of $50 \mu\text{g}/\text{mL}$, the tested compounds could rupture the cell walls of all three bacteria, resulting in membrane damage and ultimately leading to bacterial death (Figure 6). In accordance with the above-mentioned results, compounds that exhibit good antibacterial activity may have a significant tendency to show better antifouling activity.

Antifouling Studies. All the synthesized compounds (S1–S12) were evaluated for their antifouling activities by comparing with blank, control (CP), butanolide, and Cu_2O as reference paints (RP) since they have been reported to exhibit relevant biological activities. The fouling cover percentage was calculated after 60 days of sea exposure. A few organisms settled and can be seen with the naked eye on the control paint, mainly the calcareous macro fouler *Balanus Amphitrite*. Paints S1–S6 showed no significant differences with CP, while S7, S9, S8, S10, S11, and S12 showed very good differences in the settlement of *B. amphitrite* when compared with the control paint (Figure 7). As previously mentioned,

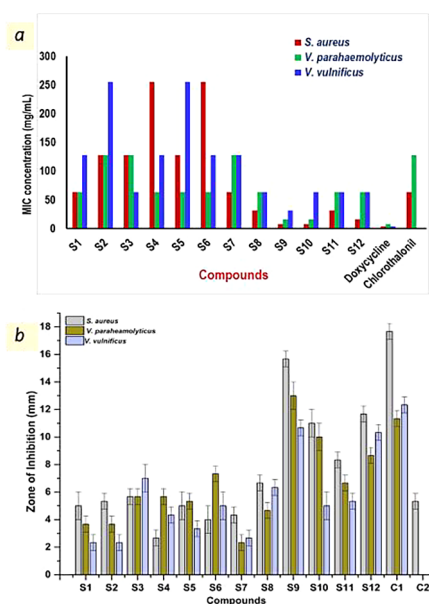


Figure 5. (a) MIC of tested compounds S1–S12. (b) Zone of inhibition of tested compounds S1–S12.

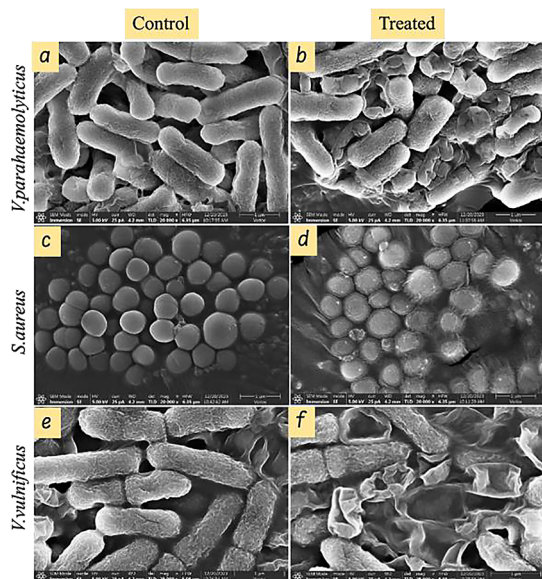


Figure 6. Scanning electron microscopy (SEM) images of *V. parahaemolyticus* (a) control and (b) treated with S9; *S. aureus* (c) control and (d) treated with S10; *V. vulnificus* (e) control and (f) treated with S12.

1,2,3 triazolyl quinolinones showed better antibiofouling activity compared to the control group among the investigated compounds. A small amount of seaweed and rare large-scale marine organisms, such as barnacles and mussels, adhered to the coating surfaces after 30 days of submersion in seawater, indicating that these compounds have good antifouling properties and could find application in the boat and marine industries. Conversely, the blank plate showed significant fouling by marine species as the mussels covered its surface and barnacles or oysters grew on it. Upon completion of 60 days of marine field testing, certain fouling organisms were found to be attached to the tested coating surface. On the other hand, a thick layer of mussels and barnacles covered the blank panel surface. The antifouling properties of the coatings are

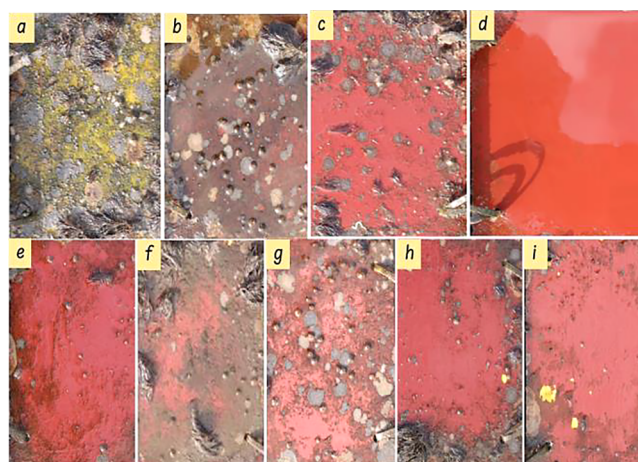


Figure 7. Coated panels after 60 days of exposure at the South China Sea; (a) blank; (b) CP, control paint; (c) butenolide paint; (d) Cu₂O paint; (e) S7; (f) S8; (g) S9; (h) S10; (i) S12 tested plates.

significantly enhanced by 1,2,3 triazolyl quinolinones and BIT-oxobutanoate.

As of now, no research has been submitted on the usage of 1,2,4-triazolo-quinolinones in AF coatings. Nevertheless, reports on BIT derivatives have been submitted, and BIT acetic acid has been synthesized using different methods and assessed for its antifouling and antialgal properties.^{62,63} In spite of these studies reporting promising performances, assessing antifouling performance using this formulation method has not been reported.⁷ Comparatively, compounds S8, S9, S10, and S11 displayed good activity when compared with the known butenolide paint. Also, S7 and S12 showed AF activity close to the most popular copper antifouling paint.

Structure–Activity Relationship Studies. Algal Inhibition. Structure–activity relationship (SAR) studies revealed that the synthesized compounds exhibited different levels of algal inhibitory activity, which can be attributed to their structural features. The 1,2,4-triazolo-pyrimidine derivative (S1) with ethyl ester enhances lipophilicity and improves membrane penetration. Also, 1,2,4-triazolo-quinazolinones (S7, S8) showed significant activity, suggesting that the additional ethyl or acetic acid groups enhance activity. Compounds S9, S11, and S12 demonstrated good inhibitory activity due to the functional importance of oxobutanoate and isothiazolone moieties.

Antibacterial and Antifouling Activities. The 1,2,4-triazolo[1,5-*a*]pyrimidine core (S3–S4) with allyl groups enhances bacterial membrane interaction, improving antibacterial action. The 3-chlorophenyl group is responsible for the high activity of compound S6, but S7 and S8 exhibit variable activities, indicating the balance of hydrophobic and hydrophilic properties. The 4-oxobutanoate group in S9 and S10 and the benzo[*d*]isothiazol-2(3*H*)-one moiety in S12 are essential for broad-spectrum antibacterial and antifouling activities due to strong interactions with bacterial enzymes or membranes.

CONCLUSIONS

The widespread use of toxic and hazardous biological agents in antifouling paints has raised worries about their impact on the maritime environment. As a result, conducting research to uncover new environmentally benign antifouling chemicals has

proven to be a challenge and a source of great interest within scientific communities. The current study investigated the in silico molecular docking studies, algal inhibition, and antibacterial and antifouling activities of compounds S1–S12. The synthesized compounds S7, S8, S9, and S12 showed significant interactions with Bap, ToxR, and RbmA receptor proteins in the in silico molecular docking interactions. Compounds S7, S8, S9, S11, and S12 showed good algal inhibition when tested against *C. curvisetus* and *Chlorella*. Significant antibacterial activity was shown against *S. aureus*, *V. parahaemolyticus*, and *V. vulnificus* strains by compounds S9, S10, and S12. Also, S7, S8, S9, S10, S11, and S12 molecules showed the most promising results in antifouling activities. SAR studies revealed that compound S8 with the additional acetic acid group showed significant activity in all the tests. Strong interactions with bacterial enzymes or membranes make the 4-oxobutanoate group in S9 and S10 and the benzo[d]isothiazol-2(3H)-one moiety in S12 important for antialgal, antibacterial, and antifouling properties.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.langmuir.4c01393>.

Data of NMR spectra of synthesized compounds S1–S12, single-crystal XRD, molecular docking analysis, and antialgal and antibacterial zone of inhibition (PDF)

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Author Contributions

S.G.: data curation, conduction of all experiments, analyzing the data, and writing the original draft. J.Y.: conceptualization, lab facility, funding, reviewing the data, editing, and supervision. M.D. and X.L.: making antifouling paints. R.L.: antialgal studies. D.W. and X.L.: single-crystal XRD analysis. A.J. and A.S.: Molecular docking studies. A.V.: antibacterial activities and SEM analysis. All authors have read and agreed to the published version of the manuscript.

Notes

The authors declare no competing financial interest.

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■ ABBREVIATIONS

CCR2	CC chemokine receptor 2
CCL2	CC chemokine ligand 2
CCR5	CC chemokine receptor 5
TLC	thin layer chromatography
BP	butenolide paint
CP	control paint
BIT	bezisothiazolone
MIC	minimum inhibitory concentration
AF	antifouling
SEM	scanning electron microscope

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